

LAB 4

Name Kaniz Fatema

ID: 20245103154

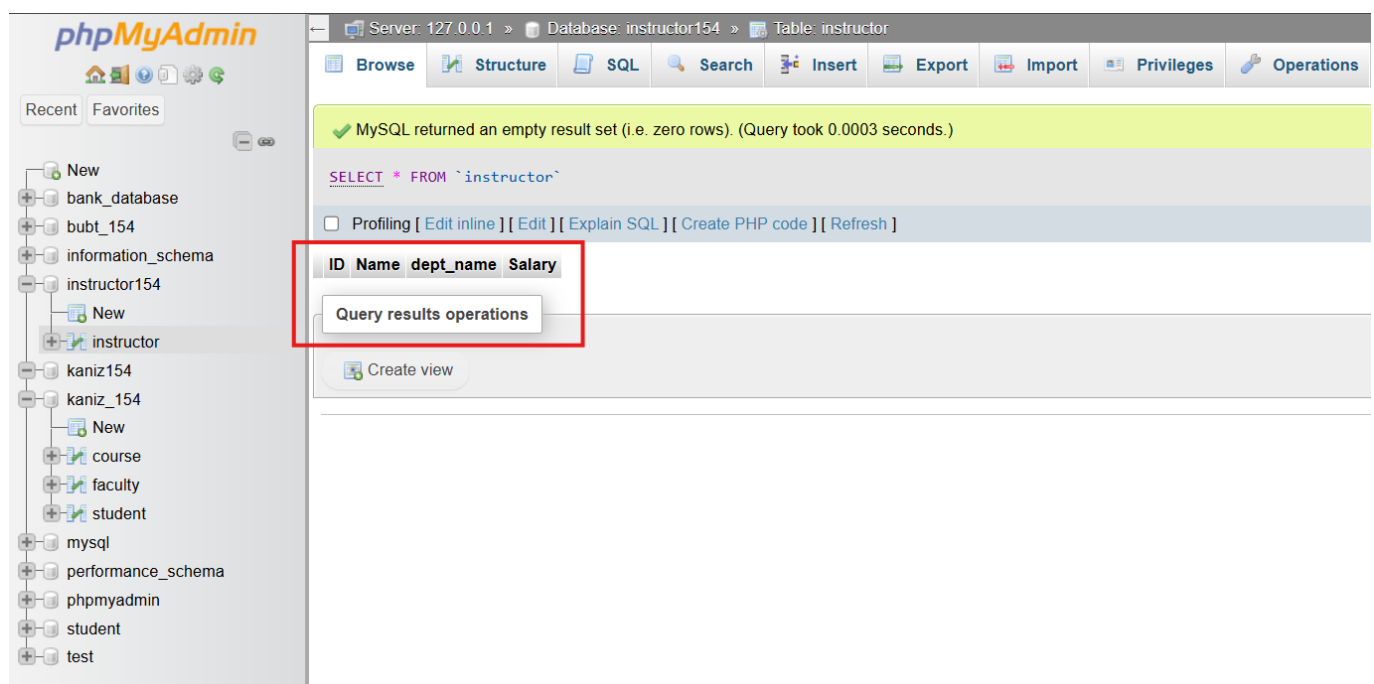
Create a DB Name "instructor154"

```
CREATE DATABASE instructor154
```

Create 1 tables in this DB named "Instructor" and "Teaches"

Instructor

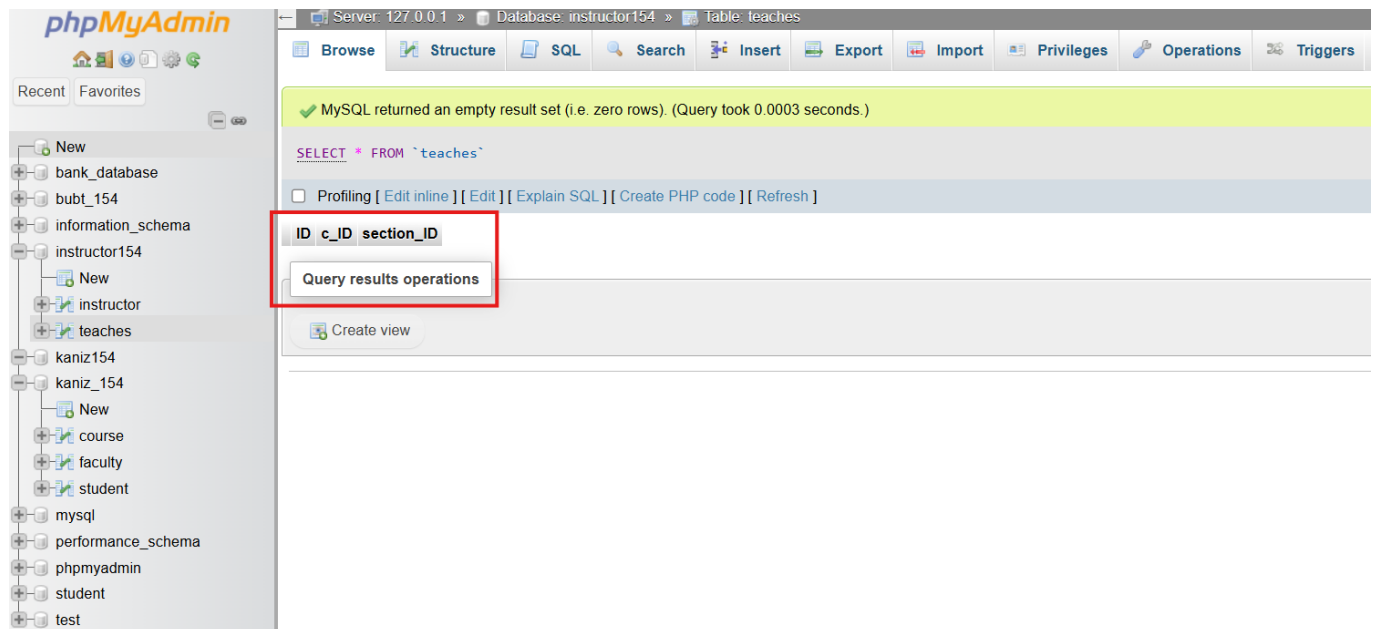
```
CREATE TABLE Instructor (ID int(8), Name varchar(32), dept_name varchar(16),  
Salary int(16))
```



The screenshot shows the phpMyAdmin interface. On the left is a sidebar with a tree view of databases and tables. The 'instructor154' database is selected, and the 'instructor' table is highlighted. The main panel shows the 'Structure' tab for the 'instructor' table. A red box highlights the table structure: columns 'ID', 'Name', 'dept_name', and 'Salary'. Below the columns is a 'Query results operations' button. The SQL query area shows 'SELECT * FROM `instructor`' and a message indicating that MySQL returned an empty result set.

Teaches

```
CREATE TABLE Teaches (ID int(8), c_ID varchar(32), section_ID int(2))
```



Insert values on "Instructor" and "Teaches"

Instructor

```
INSERT INTO instructor VALUES(10101, "Srinivasan", "CSE", 65000);
INSERT INTO instructor VALUES(12121, "Wu", "FIN", 90000);
INSERT INTO instructor VALUES(15151, "Mozart", "Music", 40000);
INSERT INTO instructor VALUES(22222, "Einstein", "Physics", 90000);
INSERT INTO instructor VALUES(32343, "Said", "History", 60000);
INSERT INTO instructor VALUES(33456, "Gold", "Physics", 87000);
INSERT INTO instructor VALUES(45565, "Katz", "CSE", 75000);
INSERT INTO instructor VALUES(58583, "Cali", "History", 62000);
INSERT INTO instructor VALUES(76543, "Singh", "FIN", 80000);
INSERT INTO instructor VALUES(76766, "Crick", "Bio", 72000);
INSERT INTO instructor VALUES(83821, "Brandt", "CSE", 92000);
INSERT INTO instructor VALUES(98345, "Kin", "EEE", 80000);
```

phpMyAdmin

Recent Favorites

New

bank_database

bubt_154

information_schema

instructor154

New

instructor

teaches

kaniz154

kaniz_154

New

course

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student

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performance_schema

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student

test

Server: 127.0.0.1 » Database: instructor154 » Table: instructor

Browse

Structure

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Triggers

Showing rows 0 - 11 (12 total, Query took 0.0001 seconds.)

SELECT * FROM `instructor`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

ID	Name	dept_name	Salary
10101	Srinivasan	CSE	65000
12121	Wu	FIN	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	90000
32343	Said	History	60000
33456	Gold	Physics	87000
45565	Katz	CSE	75000
58583	Cali	History	62000
76543	Singh	FIN	80000
76766	Crick	Bio	72000
83821	Brandt	CSE	92000
98345	Kin	EEE	80000

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Print

Copy to clipboard

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Display chart

Create view

Teaches

```
INSERT INTO teaches VALUES(10101, "CSE_101", 1);
INSERT INTO teaches VALUES(12121, "CSE_111", 2);
INSERT INTO teaches VALUES(13131, "CSE_311", 3);
```

phpMyAdmin

Recent Favorites

New

bank_database

bubt_154

information_schema

instructor154

New

instructor

teaches

kaniz154

kaniz_154

New

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Server: 127.0.0.1 » Database: instructor154 » Table: teaches

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Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 2 (3 total, Query took 0.0002 seconds.)

SELECT * FROM `teaches`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

ID	c_ID	section_ID
10101	CSE_101	1
12121	CSE_111	2
13131	CSE_311	3

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

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Create view

Q1. Perform Cartesian Product Operation between these two relations.

```
SELECT * FROM instructor, teaches
```

phpMyAdmin

Server: 127.0.0.1 » Database: instructor154 » Table: teaches

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Show all Number of rows: 25 Filter rows: Search this table

Extra options

ID	Name	dept_name	Salary	ID	c_ID	section_ID
10101	Srinivasan	CSE	65000	10101	CSE_101	1
10101	Srinivasan	CSE	65000	12121	CSE_111	2
10101	Srinivasan	CSE	65000	13131	CSE_311	3
12121	Wu	FIN	90000	10101	CSE_101	1
12121	Wu	FIN	90000	12121	CSE_111	2
12121	Wu	FIN	90000	13131	CSE_311	3
15151	Mozart	Music	40000	10101	CSE_101	1
15151	Mozart	Music	40000	12121	CSE_111	2
15151	Mozart	Music	40000	13131	CSE_311	3
22222	Einstein	Physics	90000	10101	CSE_101	1
22222	Einstein	Physics	90000	12121	CSE_111	2
22222	Einstein	Physics	90000	13131	CSE_311	3
32343	Said	History	60000	10101	CSE_101	1
32343	Said	History	60000	12121	CSE_111	2
32343	Said	History	60000	13131	CSE_311	3
33456	Gold	Physics	87000	10101	CSE_101	1
33456	Gold	Physics	87000	12121	CSE_111	2
33456	Gold	Physics	87000	13131	CSE_311	3
45565	Katz	CSE	75000	10101	CSE_101	1
45565	Katz	CSE	75000	12121	CSE_111	2
45565	Katz	CSE	75000	13131	CSE_311	3
58583	Cali	History	62000	10101	CSE_101	1
58583	Cali	History	62000	12121	CSE_111	2
58583	Cali	History	62000	13131	CSE_311	3
76543	Singh	FIN	80000	10101	CSE_101	1

Show all Number of rows: 25 Filter rows: Search this table

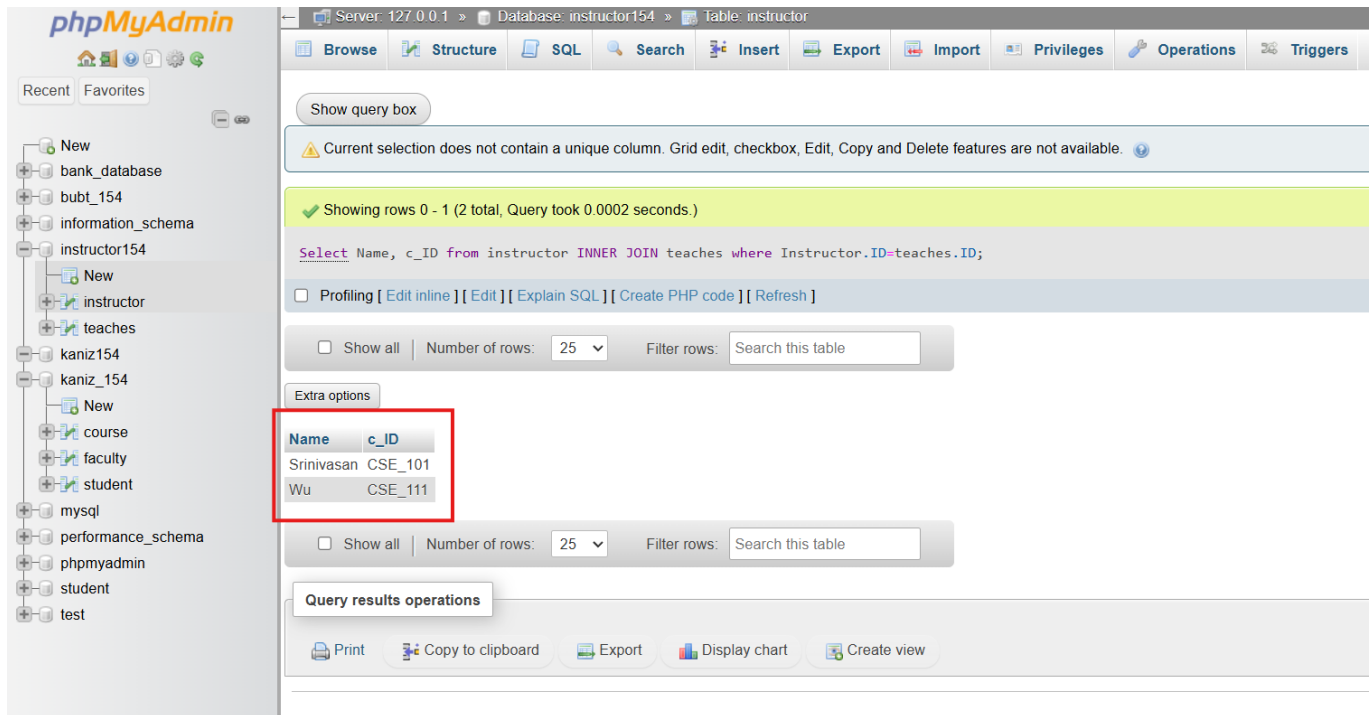
Query results operations

Q2. Find those instructors who teaches any of the courses.

```
Select Name, c_ID from instructor INNER JOIN teaches where
Instructor.ID=teaches.ID;
```

OR

```
Select Name, c_ID from instructor, teaches where Instructor.ID=teaches.ID;
```



Server: 127.0.0.1 » Database: instructor154 » Table: instructor

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 1 (2 total, Query took 0.0002 seconds.)

```
Select Name, c_ID from instructor INNER JOIN teaches where Instructor.ID=teaches.ID;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

Name	c_ID
Srinivasan	CSE_101
Wu	CSE_111

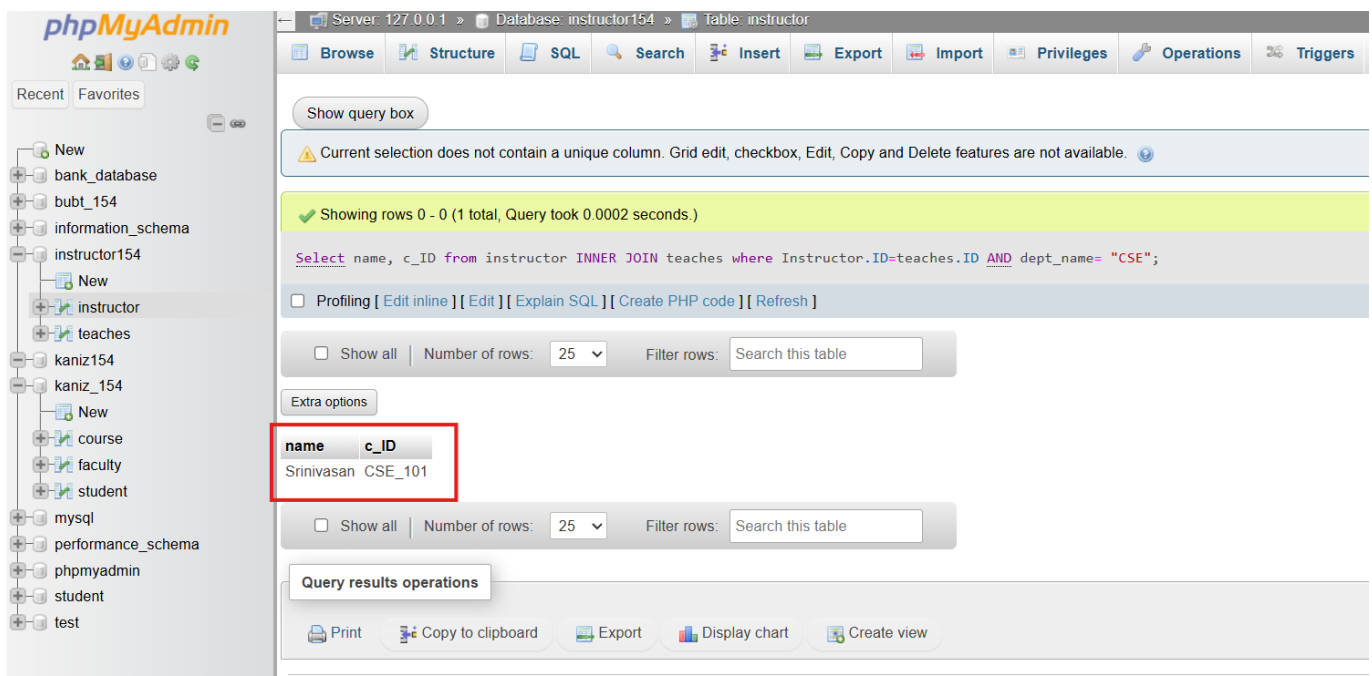
Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

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Q3. Find only instructor names and course id for instructors in the Computer Science department.

```
Select name, c_ID from instructor INNER JOIN teaches where
Instructor.ID=teaches.ID AND dept_name= "CSE";
```



Server: 127.0.0.1 » Database: instructor154 » Table: instructor

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 0 (1 total, Query took 0.0002 seconds.)

```
Select name, c_ID from instructor INNER JOIN teaches where Instructor.ID=teaches.ID AND dept_name= "CSE";
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

name	c_ID
Srinivasan	CSE_101

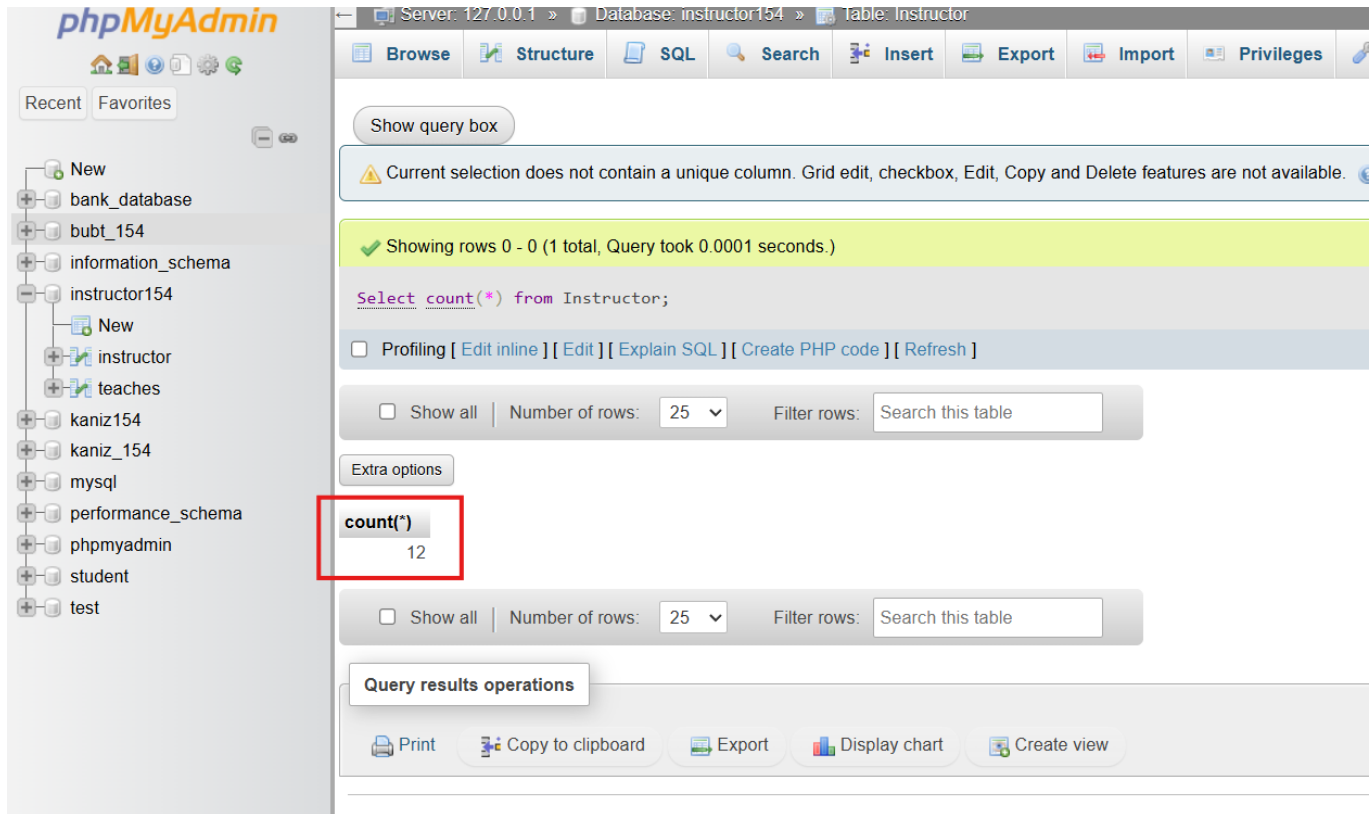
Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

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Q4. Find the total no. of tuples in "Instructor" relation.

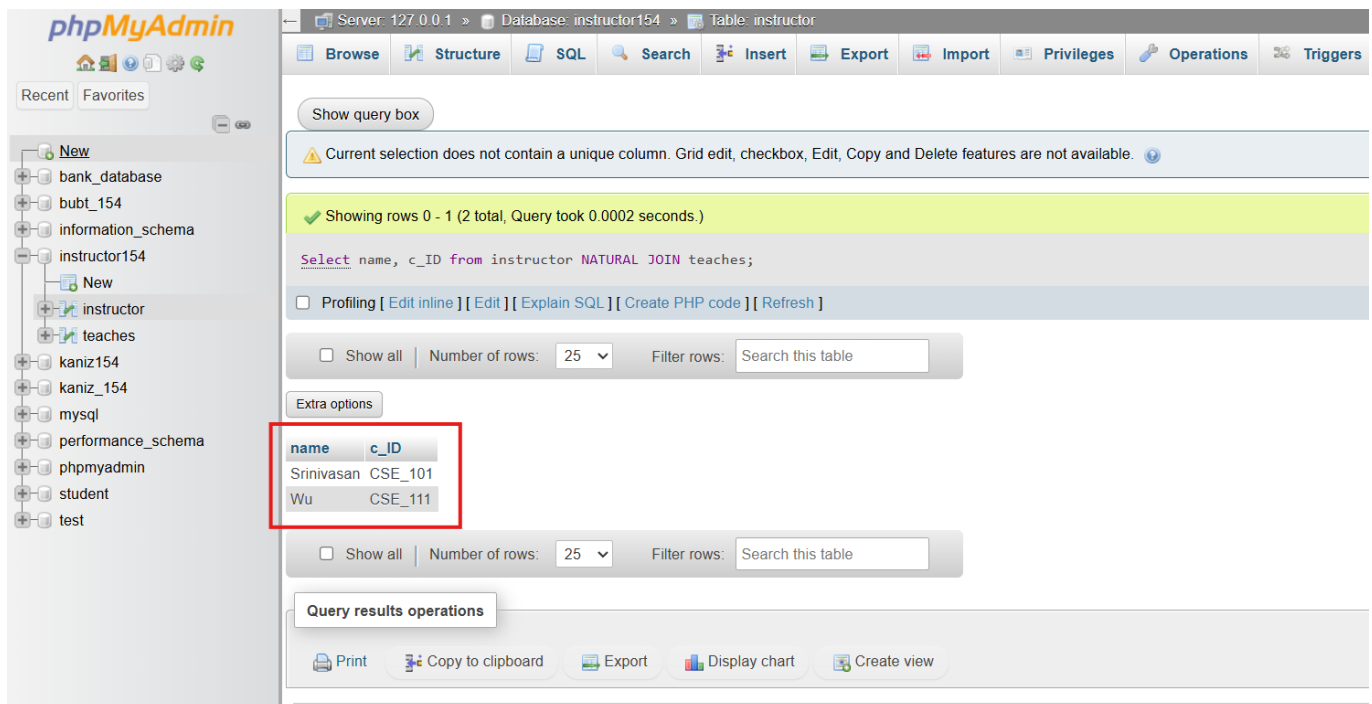
```
Select count(*) from Instructor
```



The screenshot shows the phpMyAdmin interface. On the left, the database structure is visible, including 'instructor154' and its tables 'instructor' and 'teaches'. The main panel shows the 'Instructor' table selected. A query box contains the SQL: `Select count(*) from Instructor;`. The result shows a single row with the value 12, which is highlighted with a red box. The interface also shows a warning about the lack of a unique column and various toolbars for query execution and results management.

Q5. Answer Q2 using Natural Join.

```
Select name, c_ID from instructor NATURAL JOIN teaches
```



The screenshot shows the phpMyAdmin interface with the same database structure. The main panel shows the result of the query: `Select name, c_ID from instructor NATURAL JOIN teaches;`. The result is a table with two rows: 'Srinivasan CSE_101' and 'Wu CSE_111'. This result is highlighted with a red box. The interface also shows a warning about the lack of a unique column and various toolbars for query execution and results management.

Q6. Answer Q3 using Natural Join.

```
Select name, c_ID from instructor NATURAL JOIN teaches where  
Instructor.ID=teaches.ID AND dept_name= "CSE";
```

phpMyAdmin

Recent Favorites

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bank_database

bubt_154

information_schema

instructor154

New

instructor

teaches

kaniz154

kaniz_154

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test

Server: 127.0.0.1 » Database: instructor154 » Table: instructor

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Show query box

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

✔ Showing rows 0 - 0 (1 total, Query took 0.0002 seconds.)

Select name, c_ID from instructor NATURAL JOIN teaches where Instructor.ID=teaches.ID AND dept_name= "CSE";

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

name	c_ID
Srinivasan	CSE_101

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Print

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Display chart

Create view

Q7. Perform Left Outer Join.

```
Select * from instructor NATURAL LEFT OUTER JOIN teaches;
```

phpMyAdmin

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New

bank_database

bubt_154

information_schema

instructor154

New

instructor

teaches

kaniz154

kaniz_154

mysql

performance_schema

phpmyadmin

student

test

Server: 127.0.0.1 » Database: instructor154 » Table: instructor

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✔ Showing rows 0 - 11 (12 total, Query took 0.0002 seconds.)

Select * from instructor NATURAL LEFT OUTER JOIN teaches;

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

ID	Name	dept_name	Salary	c_ID	section_ID
10101	Srinivasan	CSE	65000	CSE_101	1
12121	Wu	FIN	90000	CSE_111	2
15151	Mozart	Music	40000	NULL	NULL
22222	Einstein	Physics	90000	NULL	NULL
32343	Said	History	60000	NULL	NULL
33456	Gold	Physics	87000	NULL	NULL
45565	Katz	CSE	75000	NULL	NULL
58583	Cali	History	62000	NULL	NULL
76543	Singh	FIN	80000	NULL	NULL
76766	Crick	Bio	72000	NULL	NULL
83821	Brandt	CSE	92000	NULL	NULL
98345	Kin	EEE	80000	NULL	NULL

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

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Display chart

Create view

Q8. Perform Right Outer Join.

```
Select * from instructor NATURAL RIGHT OUTER JOIN teaches
```

phpMyAdmin

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New

bank_database

bubt_154

information_schema

instructor154

New

instructor

teaches

kaniz154

kaniz_154

mysql

performance_schema

phpmyadmin

student

test

Server: 127.0.0.1 » Database: instructor154 » Table: instructor

Browse

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Triggers

Show query box

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

✔ Showing rows 0 - 2 (3 total, Query took 0.0002 seconds.)

Select * from instructor NATURAL RIGHT OUTER JOIN teaches;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

Number of rows: 25

Filter rows: Search this table

Extra options

ID	c_ID	section_ID	Name	dept_name	Salary
10101	CSE_101	1	Srinivasan	CSE	65000
12121	CSE_111	2	Wu	FIN	90000
13131	CSE_311	3	NULL	NULL	NULL

Show all

Number of rows: 25

Filter rows: Search this table

Query results operations

Print

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